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Business Continuity Management—Identifying Relevant Processes for a Reference Model

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Abstract

Currently, a standardized process reference model specifically tailored for the business continuity management system (BCMS) is absent. Moreover, BCMS processes have not been a primary focus of ongoing research endeavors. This paper aims to fill this research gap by presenting findings from a process mapping study concerning BCMS processes within the most prominent and widely acknowledged standards for business continuity management, alongside insights gleaned from expert interviews. The authors propose a collection of BCMS processes that should comprise a BCMS process reference model intended for implementation at a maturity level tailored to individual organizational needs. It aims to strengthen the resilience of organizations to cyber threats and to optimize the processes for effective management within the disaster management cycle. The study identifies and maps the necessary processes required to build a comprehensive BCMS model. These processes include, among others, risk assessment, business impact analysis, the development of BC strategies and solutions, the creation of BC plans and procedures, incident and emergency management, and periodic reviews and exercises. The relevance of these processes was validated through expert interviews, making a clear distinction between core, management, and support processes.

Keywords: risk management; supply chain management; business continuity management; process reference models

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1. Introduction

Many organizations demonstrate a lack of preparedness and often fail to recognize the criticality of continuity [1]. According to [2,3], business continuity is "the capability of an organization to sustain the delivery of products or services at acceptable predefined levels following a disruptive incident". Ref. [4] asserts that every organization faces risks that impede its ability to achieve objectives, and severe events can threaten its existence. Hence, ensuring continuing existence is foundational for organizational goal attainment [5]. Consequently, organizations must proactively plan and execute measures to address potential threats to their existence [6] within a structured BCMS.

Setting up a BCMS is particularly important for cybersecurity in critical infrastructures like the energy sector. A non-systemic review in [7] suggests adding phases to BCM to enhance cybersecurity.

Also, the information security control 5.30 "ICT readiness for business continuity" of annex A of ISO/IEC 27001:2022 [7] can be addressed by implementing a BCMS. Given the significance of maintaining business continuity, standards like ISO 22301:2019 [2] specify the criteria for a thorough business continuity management system. These requirements cover planning, implementation, operation, and ongoing monitoring and improvement of a process-driven BCMS [2].

As ISO 22301 is requirements-oriented and does not focus on the operation of a BCMS, it does not contain a process reference model of a BCMS. Organizations often struggle with performing a checklist approach regarding the BCMS requirements because it does not necessarily lead to a sustainable operated management system. Further guidance on how to operate a BCMS is demanded. Similar to other ISO management system standards, ISO 22301 refers to a process consisting of leadership, planning, support, operation, performance evaluation, and improvement [2,8] which emphasizes the necessity of process orientation as well as integration of the planning of operations and the constant checking of planning-compliant implementation.

Elements of the BCMS are described in the form of requirements [2], best practice controls and guidelines in standards like ISO/TS 22332:2021 [9], or in the form of single processes, such as the business impact analysis process within ISO/TS 22317:2021 [10]. Studies conducted in research literature describe how to adopt a BCM within data center environments [11] or in the Internet of Things [1].

Also, a systematic literature analysis was conducted by [12] to determine a list of BCMS processes. The authors concluded that the resulting process list may be incomplete, necessitating validation against existing standards to bridge academic and business communities effectively.

Thus, developing a comprehensive BCMS core process reference model, informed by standards, and scientifically verified through expert questionnaires or interviews, becomes paramount. Nevertheless, integrating current BCMS research with established standards to develop a comprehensive, cohesive, and practical process reference model remains an ongoing challenge. Unlike information security management [13], no process reference model for business continuity management is derived from the agreed standards available. Therefore, leveraging established standards relevant to business continuity management becomes crucial to avoid redundancy while emphasizing synergies within existing management concepts. Further relevant standards or standards related to business continuity management are shown in Figure 1. Moreover, ISO 22301:2019 [2], ISO 22313:2020 [3], and ISO 22332:2021 [9] contain business continuity controls, which also partially contribute to implementing processes. For most organizations, there is no alternative to process-oriented management.

Process management addresses specific problems and serves as a platform for capitalization on new opportunities [14], a critical aspect in contemporary information-processing societies.

Hence, the primary objective of this study is to develop a detailed yet generic BCMS core process reference model that is readily adaptable and implementable across all organizations.

Considering the process reference model for information security management delineated in ISO/IEC TS 27022:2021 [13], a corresponding model for business continuity management would facilitate the effective management of business continuity. It also assists in process-oriented integration of information security management, business continuity management, and additional management systems, such as information privacy management, into an integrated management system, as expounded in the ISO Handbook "The Integrated Use of Management System Standards (IUMSS)" [15]. Research initiatives should therefore adopt a process-oriented approach, emphasizing the integration of business conti-

nuity management with existing management systems. The recommended BCMS process reference model must clearly distinguish among core processes, supporting processes, management processes, as well as the technical and organizational controls—referred to as business continuity solutions—that are regulated by BCMS protocols. Alignments with strategic goals and objectives, adaptations to individual requirements and framing conditions, as well as cost-efficiency are fundamental components of an effective management system [16]. A detailed reference model outlining BCMS processes—including inputs, outputs, interfaces, and interactions at the activity level—facilitates optimal coordination of these processes. This study addresses an existing research gap by presenting findings derived from process mapping and expert interviews focused on BCMS processes.



Figure 1. Relevant standards or standards related to business continuity management.

Both approaches aim to elucidate the research question: "Which elements does the agreed BCMS process reference model consist of?". This is decomposed into the sub-questions as illustrated in Figure 2.

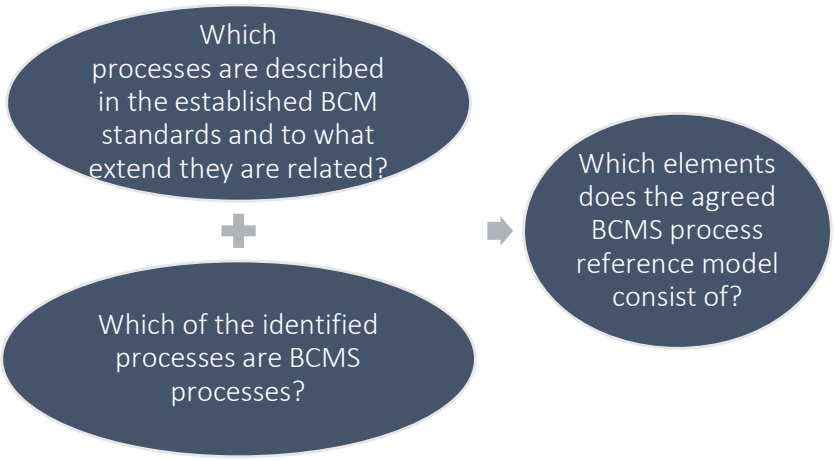


Figure 2. Research questions.

To the best of the authors' knowledge, no mapping focused on processes mentioned in those standards is available. The same applies to expert interviews identifying BCMS processes. The research findings from both methodologies have enabled the authors to formulate and present a consensus-based BCMS process reference model, thereby enriching the existing body of knowledge. This outcome comprises a comprehensive set of BCMS processes, including process descriptions (process profiles) and process flow charts, though

the latter are omitted here to maintain the scope of the paper. The BCMS processes outlined in the reference model should be implemented at a maturity level suitable to each organization's specific context. This methodological approach supports practitioners in managing business continuity with greater efficiency and effectiveness, aligning with the primary goal of the proposed reference model.

The remainder of this paper is organized as follows: Section 2 gives an overview of processes, different process types, process reference models, and the relevant ISO standards. Section 3 also explains the methods used to analyze business continuity management standards. Section 4 outlines the results of these analyses and discusses expert interviews that helped confirm the core BCMS processes, finishing with the established BCMS process reference model. The paper concludes in Section 5 with a summary and suggestions for future work.

2. Background Concepts

2.1. Processes and Process Reference Models

Organizations need to identify and manage many activities to function effectively and efficiently. Activities utilizing resources must be managed to convert inputs into outputs through a series of interconnected or interacting activities, also called processes [16]. In essence, a process represents a cohesive set of interrelated or interacting activities to transform inputs into outputs [17]. This is the objective of the process. A business process constitutes a procedure relevant to the adding of value to an organization [18]. The term "process" derives from the Latin term "procedure," which signifies going ahead and going forward.

Processes, unlike projects, can include, consist of, or start other processes, and are often performed repeatedly.

Processes frequently involve collaboration between different departments and play an integral role in an organization's operations. They can be classified into core processes, management processes, and supporting processes [18]. Core processes generate tangible and immediate value for customers, leveraging the organization's key competencies. Management processes establish organizational objectives and monitor their attainment at both the core process and enterprise levels, including functions such as project management, quality assurance, security, risk management, and strategic planning. Supporting processes facilitate and administer essential resources, operating indirectly to underpin core and management processes without directly contributing to customer value. Common examples of supporting processes are human resource management, financial management, and IT management.

According to [12], "Process Reference Model aids the organizations by clearly defining the processes for a given scope". Stated differently, it represents the fundamental framework of a management system. Furthermore, ISO/IEC TS 27022 states, "A process reference model is a model comprising definitions of processes described in terms of process purpose and outputs, together with an architecture describing the relationships between the processes" [13]. Finally, ISO/IEC 33004 [19] stipulates, "The purpose of a process reference model is to define a set of processes that collectively can support the primary aims of a community of interest. A process reference model provides the basis for one or more process assessment models". Criteria for process reference models defined in ISO/IEC 33004 are as follows:

1. A process reference model is mandated to encompass a declaration explaining the domain it pertains to.
2. It is imperative for a process reference model to explicate the relationship between itself and the context in which it is intended for use.

3. A process reference model shall contain process descriptions, meeting the following requirements within the scope of the process reference model:
    a. A process shall be described in terms of its purpose and process outcomes.
    b. The described set of process outcomes shall be necessary and sufficient to achieve the purpose of the process.
    c. Process descriptions shall not contain or imply aspects of the process quality characteristic beyond the basic level of any relevant process measurement framework conformant with ISO/IEC 33003:2015 [20].
    d. A process outcome describes one of the following: production of an artifact; a significant change in state; meeting of specified constraints, e.g., requirements, goals, etc.
4. A process reference model should delineate the interconnections among the processes outlined within it.
5. The process reference model shall document the community of interest it represents, and the actions taken to achieve consensus within that community of interest.
6. Each process delineated within a process reference model must possess unique process descriptions and identifications.

2.2. ISO Standards for Business Continuity Management

In the following, relevant ISO standards for business continuity management are described. The standards ISO 22301:2019 [2], ISO/TS 22317:2021 [10], and ISO 22300:2025 [21] are especially appropriate for developing a process reference model for BCMS, as they contain processes or hints about the processes of a BCMS.

- ISO 22300

As the basic standard of the ISO 22300 standards family, ISO 22300:2025—Security and resilience—Vocabulary [21] defines terms used in security and resilience standards. This standard concisely defines relevant terms, including those pertinent to BCMS processes. Therefore, the terms specified within ISO 22300 serve as a common language to identify and describe BCMS processes.

- ISO 22301

ISO 22301:2019—Security and resilience—Business continuity management systems—Requirements [2] specifies the structure and requirements of a business continuity management system. Compliant with ISO Directives Part 1 Annex SL [8], ISO 22301 stipulates that a BCMS includes management processes relating to policy, planning, implementation and operation, performance assessment, management review, and continual improvement. Unlike ISO/IEC 27001 [7], ISO 22301 does not incorporate an annex specifying business continuity controls or best practices. This is described in the specific standard—ISO/TS 22332—Security and resilience—Business continuity management systems—Guidelines for developing business continuity procedures [9]. This technical specification is complementary to ISO 22301. It is augmented with supplements, as neither ISO 22301 nor ISO 22313 include sufficient details regarding the methods to create BC procedures, how to structure the procedures in the form of plans, and the purpose of such procedures that outline how to recover an activity.

- ISO 22313

ISO 22313:2020—Security and resilience—Business continuity management systems—Guidance [3] provides guidance and recommendations on the requirements of ISO 22301 to explain and clarify the meaning and purpose of the requirements. Furthermore, it provides an overview of BCMS processes, especially those outlined in Clause 8, “Operation”, which

describes the steps needed to establish business continuity. In addition, Clause 4.4 emphasizes the organization's responsibility to define, implement, and sustain BCMS processes, taking into account their order, interactions, inputs, outputs, resources, and assigned roles.

- ISO/TS 22317

ISO/TS 22317:2021—Societal security—Business continuity management systems—Guidelines for Business Impact Analysis (BIA) [10] includes suggestions on how to design and implement the business impact analysis process as part of a BCMS.

- Cybersecurity and Risk Management Facets of BCM

Clause 4.1 of ISO 22301 considers the organization and its context. Within this context, cybersecurity threats are gaining importance. A poignant example are critical infrastructures, where increasing digitalization of network operations of power and gas transmission networks have made these networks more vulnerable to cybersecurity attacks [22]. Clause 8.2 of ISO 22301 relates to business impact analysis and risk assessment, which also includes cybersecurity risks and their assessment. Similarly, Clause 8.3 deals with business continuity strategies which should be identified. These strategies include specific options for the periods before, during and after an outage, and are particularly related to outages caused by cybersecurity attacks.

3. Methods for the Analysis of Business Continuity Management Standards

To answer the first research question "Which processes are described in the established BCM standards and to what extend they are related?", the approach of how to analyze business continuity management (BCM) standards is to analyze ISO 22301:2019 standard statements on processes for the purpose of comparing them, with respect to the degree of coverage and relationship with specific processes, to ISO 22300, ISO 22313, and ISO 22317 in favoring reuse, from the viewpoint of process management and management in general, and in the context of organizations interested in planning, implementing, and operating process-based business continuity management systems.

To accomplish this goal, it is essential to harmonize various process reference models. A systematic, step-by-step method for aligning different process reference models—which can be applied to any selected models—was described by [23]. As outlined in [23], this harmonization strategy is based on the process and framework for facilitating multi-model harmonization as described in [24]. According to [23], the harmonization framework is generally defined as follows:

- A guideline for determining the harmonization goals, based on the strategic plan and goals defined in the organization's mission.
- A harmonization process for driving multi-model harmonization, with which to manage and lead the harmonization of models step by step.
- A harmonization ontology, which presents the terms, concepts, and relationships for supporting the harmonization models, and
- A set of techniques and methods, which facilitates the configuration and definition of the harmonization strategies. The harmonization strategy is the work product resulting from the implementation of the harmonization process.

A theoretical comparison process is employed as the harmonization process, considering that mapping is widely utilized for harmonizing models [25]. This procedure involves systematically comparing and aligning multiple models to ensure the reliability and robustness of the resulting outcomes. The theoretical comparison initially focused on the ISO 22301 standard, which is widely considered the most critical business continuity management standard [23,24]. The result of this comparison is a table (Result of Comparison) that outlines how the models relate to one another regarding the BCM processes being

evaluated. For the purpose of analyzing the identified BCM standards, an adaptation of the Models and Standards Similarity Study method by [26] was used. The method—as it was adapted for the objectives of this study—is shown in Figure 3.



Figure 3. Study method.

Since all of the standards reviewed are international and can be applied to any organization regardless of size, goals, business model, or location, the details about BCMS processes are general. As a result, these standards were analyzed using a similar level of detail. Rather than developing an extensive correspondence template, a process profile template was established. Relevant information from the standards was integrated into these process templates. In conformance with ISO/IEC 33004 [19], the process profile templates included the following information: process name, process category (management, core, support), brief description, objective/purpose, input, outcomes, activities/functions, example metrics and informative references. The terms "matching" and "mapping" have two different definitions. Matching identifies two semantically related processes [27]. Mapping refers to the combination of the standards/processes [26]. If represented (analogy) in two or more standards employing the same or different terms, the processes are semantically related.

The criteria for determining analogy are characterized by implicit rules, facilitating knowledge mapping from a base domain (ISO 22301) to a target domain. The matching process involves posing the following questions:

1. Is there any information about BCMS processes in the other standards related to BCMS processes of the reference standard (ISO 22301)?

2. What is the additional information that could help to carry out the BCMS process of the reference standard?
3. Is there any information about possible additional BCMS processes in the other standards?
4. What is this information/what is the possible additional BCMS process?

Furthermore, a comparison scale has been defined and utilized (see Table 1). It contains the elements based on the scale presented by [25]. This scale categorizes processes into four distinct levels: The process is strongly related (S) if the process is specially named in the standards, has the same process objectives, and contains the same process steps. The process is partially related (P) if the process is not specially named; however, one or more requirements in the standard led to the implementation of the process defined in another standard, or the process was named but not described in more detail. The process is weakly related (W) if the process is not specially named, but there is a process or a process concept that can/should be adapted in a BCMS. Finally, the process is not related (N) if no relationship can be identified.

Table 1. Matrix of analyzed standards and contained BCMS processes (S—strongly related, P—partially related, W—weakly related and N—not related).

Process/Standard	ISO 22301:2019	ISO 22300:2025	ISO 22313:2020	ISO/TS 22317:2021
BCMS planning process	S	P	S	N
Business continuity governance process	S	S	S	N
Business continuity policy management process	S	S	S	N
Requirements management process	S	N	S	N
Business impact assessment and criticality analysis process	S	S	S	S
Risk assessment process (for risks of disruption)	S	S	S	S
Process of determining and selecting BC strategies and solutions	S	P	S	P
Solution implementation management process	S	P	S	P
Process of developing BC plans and procedures	S	P	S	P
Incident and emergency handling process	S	S	S	N
Warning and communication process	S	P	S	N
Recovery process	S	S	S	N
Process of exercising BC plans and procedures	S	S	S	N
Awareness, competence, and training process	S	N	P	N
Supply chain management process	S	S	S	N
Business continuity changes management process	S	N	S	P
Internal audit process	S	S	S	P
Performance evaluation process	S	S	S	P
Continual improvement process	S	S	S	P
Process of controlling documented information	S	S	S	N
Resource management process	S	P	S	N
Consultation and communication process	S	S	S	N
Customer relationship management process	S	P	S	N

After identifying semantically related processes, they are combined into an integrated process reference model through mapping [27]. Matching and mapping are established methods in scientific knowledge comparison [27], primarily employed for comparing and merging different ontologies [27,28]. In the matching step, the comparison is performed through an iterative and incremental procedure.

The process is iterative because the comparison (analysis and determination of the relationship between the relevant standards) is executed comprehensively on one BCMS process before proceeding to others. It is also incremental as the comparison outcome (i.e., the final product of the theoretical comparison process) grows and evolves with each iteration until it reaches its final form. This iterative and incremental approach was deemed necessary to address the complexity inherent in comparing entities of low-level abstraction.

For process identification, the following method was employed (results are summarized in a mapping table in the subsequent section): initially, ISO 22301 was analyzed regarding the mentioned processes. The result of this task is documented in the next section. ISO 22300, ISO 22313, and ISO 22317 were analyzed (matching) regarding BCMS processes already identified in ISO 22301, as well as potential additional BCM processes. A matching table regarding the possible BCMS processes was created for the other standards. The result of this task is documented in the next section.

4. Results

4.1. Results of the Analysis of Business Continuity Management Standards

The analyzed standards do not provide a BCMS process reference model, which includes a detailed description of BCMS processes, their inputs, outputs, and interfaces. Table 1 presents a matrix illustrating the analyzed standards alongside the identified potential BCMS processes. This analysis addresses the primary research question: "What processes in the established business continuity management standards are described, and to what extent are they related?"

Establishing the criteria for categorizing processes as BCMS core processes is imperative to address the second research question. These criteria must be formulated to identify BCMS core processes, as there is currently no universally accepted set of criteria for this purpose. The authors addressed this gap by conducting the initial phase of a study [29], utilizing an expert questionnaire to identify relevant criteria for ISMS processes. These criteria were subsequently adapted for BCMS process identification. As previously defined, core processes deliver evident and direct customer value and stem from an organization's core competencies. Within the context of BCMS, a core process constitutes the core competency of the BCMS and delivers apparent and direct value to stakeholders, typically the implementing organization or its management.

Processes in general, and BCMS processes, in particular, necessitate the consideration of regular tasks, their interrelationships and interactions, as well as the transformation of inputs into outputs, resulting in input–transformation–output processes [30]. Consequently, the following BCMS core processes are defined as operational processes (mainly—but not limited to—part of the DO and CHECK steps of the PDCA cycle) that require regular repetition. Tasks undertaken during the planning, development, or improvement of the BCMS in the Plan and Act stages of the PDCA cycle are analyzed to identify elements requiring regular repetition and integration into operational processes.

Furthermore, a criterion for categorizing a process as a BCMS core process is that the business continuity manager owns the process (core competency of the BCMS). This means that the business continuity manager is accountable for the whole process. As the process owner, they define the process's objectives derived from the BCMS's objectives, which the senior management will define. The business continuity manager is also responsible for the process design [31].

In many instances, particularly within small organizations, the business continuity manager may also serve as the process manager responsible for process operation. This will also be used as a criterion for BCMS core processes. In summary, the authors identified

the following fundamental criteria for BCMS core processes before initiating the study, which are shown in Figure 4.

Criteria 1 – Regularity – interrelated and interacting tasks are repeated on a regular basis.
Criteria 2 – Transformation – inputs are transformed into outputs.
Criteria 3 – Operationally – process is carried out while operating the BCMS.
Criteria 4 – Accountability/responsibility – business continuity manager is the process owner or process manager and the process is a core competency of the BCMS.
Criteria 5 – Value generating – delivers apparent and direct value to the stakeholder.

Figure 4. Criteria for BCMS core processes.

By applying the defined criteria for BCMS core processes to the mapping results, a list of BCMS core processes was derived.

Consequently, all processes outlined in Table 1 were confirmed as BCMS processes except for the BCMS planning process, which lacks regularity and operationality. Therefore, the BCMS planning process is categorized as a project. Similarly, the business continuity governance process was classified as a management process as it lacks operationality. The following processes are categorized as support processes as they deliver no direct value to the stakeholders: process of controlling documented information, resource management process, consultation and communication process, and customer relationship management process. All other processes from Table 1 were confirmed as BCMS core processes.

A possible integration of the processes into the disaster management cycle (mitigation, preparedness, response, recovery) can be achieved by mapping on them the processes from the general PDCA management system cycle, e.g., PLAN will roughly correspond to mitigation, DO and CHECK will encompass preparedness, and, finally, ACT will mirror response and recovery

4.2. Expert Interview Verification

To validate or refute the identified BCMS core processes or add any potentially missing BCMS core processes, the authors of this paper conducted expert interviews.

During these interviews, 47 experts were presented with a questionnaire soliciting their identification of BCMS core processes. The questionnaire included a predefined set of BCMS processes delineated in Table 1. Only the process names contained in the first row were provided. Also, an initial differentiation was provided in core, management, and support processes. Additionally, experts were allowed to suggest additional processes in the questionnaire. Notably, processes not initially identified as BCMS core processes were integrated, including the capability assessment process and the process of ensuring sustained quality of products and services. The panel of experts comprised 47 German professionals specializing in business continuity, of which 39 responded to the questionnaire. These experts included 14 business continuity managers (10 from private companies and four from public administration) and 25 expert/senior consultants for business continuity. The results of the interviews for identifying BCMS core processes were categorized into three levels for discussion, as depicted in Figure 5.

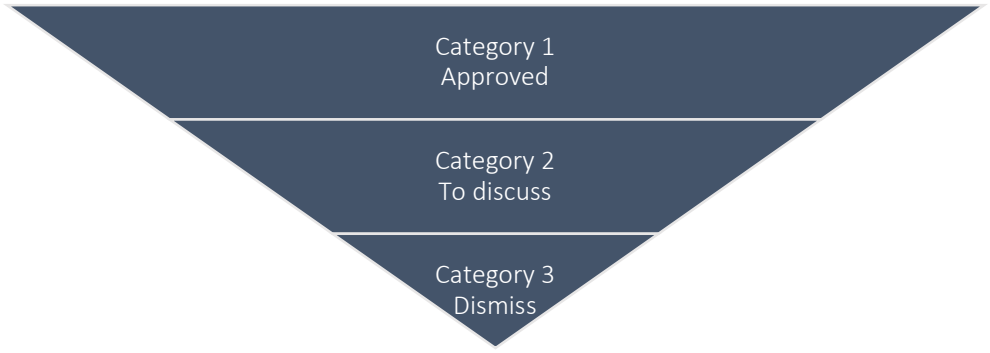


Figure 5. Categories used in the interview.

According to [32,33], the 80% threshold for consensus is commonly used and generally perceived as fair.

Category 1 processes represent those clearly identified as BCMS core processes by 80% or more of the experts. Category 2 processes were not clearly identified as BCMS core processes, recognized by 20% to 80% of the experts. Category 3 processes were identified as not being BCMS core processes, recognized by less than 20% of the experts. Despite the potential bias introduced by the predefined set of processes in the questionnaire, experts had the liberty to propose additional processes, none of which were suggested by any expert.

Furthermore, the questionnaire included two processes that were not initially identified as BCMS core processes, namely, the capability assessment process and the process of ensuring sustained quality of products and services. Consequently, the degree of bias is deemed acceptable. The results are as follows:

Category 1 processes, which were confirmed as BCMS core processes by most experts, are detailed in Table 2.

Table 2. Category 1 (approved) processes.

Named Process	How Often Was the Process Named?
Business continuity governance process	35 (90%)
Business continuity policy management process	34 (87%)
Business impact assessment and criticality analysis process	39 (100%)
Risk assessment process (for risks of disruption)	35 (90%)
Incident and emergency handling process	39 (100%)
Warning and communication process	36 (92%)
Recovery process	39 (100%)
Process of exercising BC plans and procedures	39 (100%)
Awareness, competence, and training process	37 (95%)
Supply chain management process	35 (90%)
Business continuity changes management process	37 (95%)
Internal audit process	38 (97%)
Performance evaluation process	34 (87%)
Continual improvement process	35 (90%)
Process of controlling documented information	38 (97%)
Resource management process	39 (100%)

Notably, all these processes were previously acknowledged as BCMS core processes in the mapping of ISO 22300:2025 [21], ISO 22301:2019 [2], ISO 22313:2020 [3], and ISO/TS 22317:2021 [10]. Category 2 processes, which did not receive clear consensus as BCMS core processes, are outlined in Table 3.

Table 3. Category 2 (to be discussed) processes.

Named Process	How Often Was the Process Named?
Requirements management process	29 (74%)
Process of determining and selecting BC strategies and solutions	29 (74%)
Solution implementation management process	27 (69%)
Process of developing BC plans and procedures	27 (69%)
Consultation and communication process	29 (74%)

Only 74% of the experts surveyed identified the requirements management process as a BCMS core process, slightly below the 80% consensus for other clearly identified BCMS processes. Nevertheless, it is noteworthy that not all experts acknowledged this process as integral to the BCMS core process, which may indicate a lack of awareness regarding the pivotal role of identifying business continuity requirements for the success of the BCMS. Failure to ascertain these requirements invariably leads to inefficient and ineffective management practices. Due to not meeting the 80% threshold, ISO 22301 has been double-checked regarding this process. This process has been clearly identified as part of the BCMS, based on ISO 22301, as it is a requirement in Chapter 4.2.2 of ISO 22301 stating that "The organization shall implement and maintain a process to identify, have access to, and assess the applicable legal and regulatory requirements related to the continuity of its products and services, processes, activities and resources, as well as the interests of relevant interested parties." So, even if no consensus was reached within the interviewed experts, and despite the lower percentage attributed to this process in corresponding studies as an IPMS/ISMS core process, the authors of this study affirm its status as a core process within the BCMS framework. This affirmation stems from its explicit identification in [2].

In a similar vein, just 74% of respondents considered the process of determining and choosing BC strategies and solutions as core BCMS processes, while only 69% identified developing BC plans and procedures as such. Both figures are slightly below the 80% benchmark used to clearly define BCMS processes. It is unexpected, considering that both processes effectively manage recognized business continuity risks. What stands out even more is that every respondent (100%) identified the implementation of BC plans and procedures as a fundamental process within the BCMS. This prompts a reconsideration of classifying the development of business continuity plans and procedures as non-core activities, given the logical progression from plan formulation to implementation. Due to not meeting the 80% threshold, ISO 22301 also has been double-checked regarding these processes. These processes have been clearly identified as a part of the BCMS based on ISO 22301, as it is a requirement in chapters 8.3 and 8.4 of ISO 22301, which is also reflected in the chapter titles "Business continuity strategies and solutions" and "Business continuity plans and procedures". So, even if no consensus was reached within the interviewed experts, the authors assert the presence of both processes as core components of the BCMS, based on their explicit definition in [2,21].

The solution implementation management process was not recognized as a BCMS core process, with only 69% of experts categorizing it as such. This is surprising since business continuity management typically emphasizes technical and operational redundancies. The ambiguity surrounding the development of BC plans and procedures, as well as the process for managing solution implementation, may be a contributing factor to this inconsistency. This may also account for the limited emphasis placed on the processes involved in identifying and selecting business continuity strategies and solutions, as well as in developing business continuity plans and procedures. Due to not meeting the 80% threshold, ISO 22301 has been double-checked regarding this process. This process has been clearly identified as a part of the BCMS based on ISO 22301, as it is a requirement in

chapter 8.3 of ISO 22301 stating that "The organization shall implement selected business continuity solutions so they can be activated when needed." Obviously, such an implementation needs to be managed and requires a corresponding process. So, even if no consensus was reached within the interviewed experts, the authors recognize the necessity of solution implementation within the BCMS core processes, as outlined by ISO standards [2,21].

Surprisingly, less than 74% of BCMS experts identified the consultation and communication process as a BCMS core process. A possible reason for this is insufficient understanding of the importance of consistent and structured communication with BCMS stakeholders and customers. The findings from the customer relationship management process provide strong evidence for this explanation. Nevertheless, the substantial recognition rate (92%) of the warning and communication process highlights the critical role that effective communication plays within the BCMS framework. Due to not meeting the 80% threshold, ISO 22301 has been double-checked regarding this process. This process has been clearly identified as a part of the BCMS based on ISO 22301, as it is a requirement in chapter 7.4 of ISO 22301 to establish a communication process, while ISO 22300 clarifies in chapter 3.41 stating that "communication and consultation" is required. So, even if no consensus was reached within the interviewed experts, and despite the lower recognition shown in the results of the expert survey, the authors consider the consultation and communication process to be BCMS core processes.

Category 3 processes, clearly identified as not being BCMS core processes, are delineated in Table 4.

Table 4. Category 3 (disapproved) processes.

Named Process	How Often Was the Process Named?
BCMS planning process	2 (5%)
Customer relationship management process	1 (3%)
Capability assessment process	1 (3%)
Process of ensuring sustained quality of products and services	1 (3%)

The BCMS planning process was also not categorized as a BCMS core process in the mapping of ISO 22300:2025 [21], ISO 22301:2019 [2], ISO 22313:2020 [3], and ISO/TS 22317:2021 [10]. Subsequently, through interviews, it was also confirmed that this process does not constitute a fundamental aspect of the BCMS core processes. This finding is noteworthy as it challenges the conventional understanding of BCMS planning as a process executed solely at the inception of the BCMS lifecycle. Such insights were unexpected by the authors of these interviews but aligned with analogous findings pertaining to information privacy management processes, reflecting the authors' perspective.

Interestingly, only a minority of experts (3%) recognized the customer relationship management process as a BCMS core process. This observation may stem from a lack of awareness regarding the importance of ensuring adequate customer satisfaction, balancing the benefits and costs of business continuity investments, and consistently demonstrating the value added by the BCMS or business continuity controls. Likewise, within information security management and information privacy management, investments in business continuity do not yield direct monetary returns on investment. Instead, the value added by such investments is perceptible indirectly through mitigating opportunity costs in the event of incidents or events threatening security, privacy, or continuity objectives. In light of this rationale, the authors believe the customer relationship management process should be encompassed within a BCMS process reference model, notwithstanding its limited recognition by experts. The capability assessment process was included, and sustained product and service quality assurance was implemented to mitigate bias. As anticipated,

neither of the processes was identified as part of the BCMS process. In summary, the expert interviews largely validated the BCMS core processes established through mapping relevant standards.

4.3. Concluded BCMS Processes

To ensure organizational resilience in the face of disruptive incidents, the following processes have been identified as BCMS processes as a result of a comprehensive mapping study, mapped to PDCA and the disaster management cycle, and categorized into management, core and supportive processes.

Every process listed in Table 5 is essential for making the BCMS effective, efficient, and appropriate. The business continuity governance process, in particular, ensures that the BCMS matches the goals and requirements of the stakeholders responsible for oversight.

Table 5. Overview of Concluded BCMS processes.

Process	PDCA	Disaster Management Lifecycle	Process Category
Business continuity governance process	Not mapped as no core process	Not mapped as no core process	Management
Business continuity policy management process	Plan	Mitigation	Core
Requirements management process	Plan	Mitigation	Core
Business impact assessment and criticality analysis process	Plan	Mitigation	Core
Risk assessment process (for risks of disruption)	Plan	Mitigation	Core
Process of determining and selecting BC strategies and solutions	Plan	Mitigation	Core
Process of developing BC plans and procedures	Plan	Mitigation	Core
Solution implementation management process	Do	Preparedness	Core
Incident and emergency handling process	Do	Response	Core
Process of exercising BC plans and procedures	Do	Preparedness	Core
Awareness, competence, and training process	Do	Preparedness	Core
Supply chain management process	Do	Preparedness	Core
Internal audit process	Check	Preparedness	Core
Performance evaluation process	Check	Preparedness	Core
Continual improvement process	Act	Preparedness	Core
Business continuity changes management process	Act	Preparedness	Core
Warning and communication process	Act	Response	Core
Recovery process	Act	Recovery	Core
Process of controlling documented information	Not mapped as no core process	Not mapped as no core process	Support
Resource management process	Not mapped as no core process	Not mapped as no core process	Support
Consultation and communication process	Not mapped as no core process	Not mapped as no core process	Support
Customer relationship management process	Not mapped as no core process	Not mapped as no core process	Support

The business continuity policy management process facilitates the systematic development, maintenance, and preservation of business continuity policies, standards, procedures,

and guidelines (further referred to as "BC policies"). This process also ensures that BC policies are accessible to, and clearly understood by, the intended audience.

A thorough grasp of what interested parties expect and need, as it relates to business continuity and the BCMS, is essential for reaching the system's goals. The requirements management process helps ensure this by identifying relevant legal, statutory, regulatory, and contractual requirements. These requirements inform processes such as business impact assessment and criticality analysis, risk assessment, internal audits, and supply chain management.

The business impact assessment and criticality analysis process evaluates the effects of disruptive events on an organization. This procedure involves systematic prioritization of products, services, processes, and activities, followed by comprehensive analysis, consolidation, and securing of executive approval of the BIA findings. The final result is a documented statement accompanied by a rationale outlining the organization's business continuity requirements. The risk assessment process, which focuses on identifying potential disruptions, relies on the outcomes of the business impact assessment and criticality analysis as its foundation. Following this, the likelihood of disruptive incidents within processes is evaluated. This approach results in a documented catalog of assessed risks, prioritizing threats, vulnerabilities, risk owners, consequences, business impact, likelihood, and alignment with established risk criteria. Furthermore, it includes risks associated with proposed changes, providing essential input for both communication activities and the development of business continuity plans and procedures.

After the completion of the business impact assessment and criticality analysis process, the approved business continuity requirements guide the determination and selection of suitable BC strategies and solutions, thus facilitating an efficient response and recovery from a disruptive incident. This is the objective of the process of determining and selecting BC strategies and solutions. The solution implementation management process involves initiating and verifying BC strategies, solutions, and changes that may result thereof. This process ensures that the BC strategies, solutions, and necessary changes are executed as planned. The identified BC strategies and solutions aim to reduce the likelihood of disruptions, shorten their duration, and/or mitigate their impact on the organization's products and services [2] (p. 17). The activation of certain BC strategies and solutions is triggered when required, typically in response to a disruptive event, through the execution of BC plans and procedures.

Corresponding BC plans and procedures, derived from the identified BC strategies and solutions, are developed within the process of developing BC plans and procedures.

The incident and emergency handling process defines the comprehensive approach for managing disruptive events. It includes two primary subprocesses: the warning and communication process and the recovery process. The former ensures the availability and adequate execution of internal and external communications during disruptive events, facilitating appropriate alerting of potentially affected parties and coordinating responses within the organization's BC plans and procedures. The recovery process, on the other hand, involves restoring and resuming regular business activities from the state of temporary measures that have been adopted to support normal operations during and after a disruption. The process of exercising BC plans and procedures ensures that BC strategies and solutions and their corresponding BC plans and procedures, including warning and communication procedures, are tested regularly to validate their effectiveness and prompt any necessary adjustments or enhancements.

In the process of assuring necessary awareness and competence, a business continuity awareness, training, and education program is developed and implemented to foster necessary awareness and competence among personnel. Additionally, within the broader

scope of supply chain management, outsourced services must be assessed, analyzed, and controlled regarding business continuity considerations. This is realized within the supply chain management process.

Implementing and refining BC strategies and solutions invariably entails organizational changes, which can be managed through either the general change management process or, if specific to elements of the BC management system, the dedicated change management process within the BCMS framework. The latter process oversees changes to BCMS elements and evaluates the consequences of unintended alterations, focusing squarely on the management of BCMS-related changes. The output of this process encompasses:

- Necessary changes (for the process of controlling documented information);
- Proposed and necessary changes, as well as results of changes (for and from the risk assessment process);
- Initiation of risk assessment in response to significant proposed or occurring changes; and
- The results of changes to the incident and emergency handling process or the process of exercising BC plans and procedures—as those processes initiated them.

The performance evaluation process encompasses monitoring, measurement, analysis, and evaluation of the BC strategies, solutions, plans, procedures, and the performance of the BCMS processes. Performance measurement differs from performance audit (internal audit) regarding the effectiveness and efficiency of the BCMS and implemented strategies, solutions, plans, and procedures, which are performed independently within the internal audit process. Results from the performance evaluation process, the internal audit process, and the supply chain management process are used to improve the effectiveness, efficiency, suitability, and adequacy of the BCMS and the strategies, solutions, plans, and procedures. This is realized within the continual improvement process.

The process of controlling documented information involves identifying, creating, updating, and controlling information essential for the effectiveness of the BCMS. Resources necessary for executing strategies and BCMS processes are identified, allocated, and monitored within the resource management process. Outputs from this process include documented resources for implementing and running selected strategies, as well as resources for operating core BCMS processes, along with reports on resource utilization for BCMS core processes and the customer relationship management process—reports on resource usage.

The outcomes of nearly all BCMS processes are communicated centrally to stakeholders external to the BCMS as part of the consultation and communication process. It encompasses the communication of risks and reports concerning BC management. These reports and identified requirements serve as inputs for the BC governance process. While the warning and communication process is executed during a disruptive event, the consultation and communication process represents the general mode of communication even in the absence of such an event. Within this framework, consultation denotes a bilateral exchange of informed communication between an organization and its stakeholders. Consultation is pivotal in influencing decisions and furnishing input to decision-making processes, though it does not entail joint decision-making.

In addition to the BC governance process, which serves as the interface between the BCMS and its stakeholders, operational management of customer satisfaction levels and continuous demonstration of the added value of investments in business continuity need to be realized. This is done within the customer relationship management process.

Furthermore, initiating a BCMS typically involves a BCMS planning project, where inputs such as stakeholders' vision are translated into outputs like management approval for the BCMS or the BCMS scope. Specific outputs, such as management approval and

scope definition, necessitate regular scrutiny for relevance and appropriateness. However, the project itself is primarily an initial project that is carried out once. Regular activities like renewing the management approval are also integrated into the business continuity governance processes.

The BCMS planning project constitutes a component of the planning phase in the Plan-Do-Check-Act (PDCA) cycle. It is distinct from the operational phase of the BCMS (Operation or Do phase). The BCMS design project plays a pivotal role in creating value for senior management, forming the foundation for BCMS implementation, articulating objectives for the BCMS, and ensuring alignment with the requirements of top management.

5. Discussion

In [12], the authors did a systematic literature review to "collect Business Continuity Management processes" and suggested, also, to perform an investigation of ISO 22301 to conclude a process reference model for business continuity management. So, our work is focused and therefore limited to an analysis of the relevant standards, expert interviews, and pilot implementation, rather than repeating the research of scientific articles and studies.

A comparison of the BCMS process reference model derived from our study with the process catalog proposed by [12] reveals that our proposed model encompasses the processes outlined in the process catalog, and augments them with a comprehensive validation, elucidation, and incorporation of any omitted processes. Thus, our BCMS process reference model represents an important conduit between academia and industry while fostering integration and cross-pollination between scientific inquiry and business practice.

Furthermore, this BCMS process reference model has undergone practical implementation and is currently operational within a significant public sector entity as part of a pilot initiative. Some preliminary findings from this pilot deployment include:

- Integrating BCMS and ISMS processes into a unified process model yields substantial synergy.
- An unmodified application of the BCMS process reference model is inadequate; customization of BCMS processes according to the organization's specific needs is imperative. Nevertheless, the provided model serves as a valuable starting point, directing focus toward a process-oriented perspective rather than solely on measures. This shift is particularly advantageous as it precludes the notion of perceiving business continuity as a one-off project, favoring instead a continuous process-oriented approach that better aligns with the operational requirements of a BCMS. Moreover, utilizing a BCMS process reference model as a foundational framework obviates the necessity for the implementing organization to delve into the intricacies of BCMS processes, as these are already delineated.
- In addition to customizing BCMS processes, variations in the maturity levels of implemented processes are observed.

In summary, the initial evaluation of implementing the proposed BCMS process reference model has the following advantages compared to a solution/project-oriented approach:

- Efficiency: elimination of the need for the implementing organization to conduct exhaustive research on potential BCMS processes within the ISO standards, as they are readily available within the reference model.
- Operational focus: the adoption of the BCMS process reference model redirects attention from singular solutions/projects towards a comprehensive process-oriented perspective, thereby enabling and supporting the operation of a BCMS.

The identification and alignment of BC processes improve resilience to cyber threats in critical infrastructure and digital operations in various ways. It includes a risk assessment that takes cyber risks into account and a business impact analysis that prioritizes critical infrastructures and services. The model guides the selection of BC strategies and solutions tailored to cyber risks and includes the development of BC plans and procedures to respond to cyber incidents. Incident and emergency management ensures effective communication and recovery. Regular reviews of BC plans and procedures and a process-oriented approach with continuous improvements are essential. The model is led by a business continuity manager who is responsible for process design and ensures regular repetition of BCMS core processes. The core processes of the BCMS are regularly evaluated and adjusted to ensure the effectiveness and efficiency of the system.

This model provides a comprehensive framework for organizations to strengthen their resilience to cyber threats and promotes proactive planning, regular testing, and continuous improvement.

The pilot implementation of the proposed BCMS process reference model underscores that adopting a process-oriented view of the BCMS can enhance operational focus and efficiency in planning such processes. A key finding of this pilot initiative is the reinforcement of the systemic nature of the BCMS, emphasizing the interplay between processes and the roles within the BCMS framework. Furthermore, the pilot implementation demonstrates significant synergy effects, achievable through integrating BCMS with ISMS. However, it is evident that an unaltered implementation of the reference model would be inadequate. Furthermore, current limitations of the work include the limited geographic footprint of evaluation in implementation activities that were focused mostly within Central Europe, and more specifically, Germany. Nevertheless, due to the leading role of Germany with regard to cybersecurity, business continuity, and resilience, these results can serve as an initial validation of the approach and point to a wider applicability in the future.

6. Conclusions

The identified core processes of BCMS serve as a foundational framework for organizations embarking on designing and implementing such systems. Emphasizing adjustment and cost-effectiveness is pivotal for ensuring the efficacy of a management system [34]. A comprehensive reference model delineating BCMS processes, encompassing inputs, outputs, and interfaces, facilitates an appropriate understanding of their interplay at an operational level. This reference model departs from the conventional standards-centric approach towards a pragmatic, operational orientation. By leveraging this reference model, practitioners, including business continuity managers, gain expedited insights into their respective process management responsibilities. This approach empowers business continuity professionals to prioritize the operational facets of a BCMS without succumbing to the overwhelming array of business continuity plans, controls, and measures. Additionally, adopting a process-oriented perspective allows practitioners to engage in maturity discussions on each BCMS process, thereby enabling alignment with organizational requirements. This alignment fosters more efficient and effective business continuity management, which constitutes the overarching objective of the proposed reference model.

The primary applications of this reference model ideally manifest during the planning and implementation phases of a BCMS or, alternatively, during its operational phase following initial adoption. To optimize efficiency and effectiveness, the standardization of BCMS processes is imperative, particularly within large organizations or for those with multiple or multinational sites [35].

An analysis of business continuity management standards endeavors to meet research criteria encompassing relevance, applicability, and specificity [36]. Our study adheres to these criteria as follows:

- Relevance—taking into account the increasing dependability of organizations from dealing with unforeseen events threatening the existence of the organization, an effective and efficient business continuity management is highly relevant to nearly all organizations.
- Applicability—the proposed BCMS process reference model is applicable to all organizations independent of their size, objectives, business model, location, et cetera, as the underlying international/national standards are applicable to those.
- Specificity—differentiation of a set of 22 BCMS processes supported by generalized process descriptions, process profiles, and process flow-charts is much more specific than in actual research in this field or in the included information in the underlying standards. The process flow-charts will be expanded upon in future publications.

A method for determining the requisite capability or maturity level of BCMS processes may be employed within the BCMS framework to incorporate discussions on capability or maturity. Recognizing resource constraints and the imperative of efficient utilization, it is necessary to avoid establishing and operating every BCMS process at an identical level of maturity. Therefore, the future work of the authors will consist of three steps:

Step 1: Development of a generic process reference model for management systems

Every management system shares a fundamental objective: mitigating risks within a specific domain, such as information security, privacy, quality, continuity, compliance, etc., which may impact the attainment of an organization's mission and objectives. The elements and methodologies of management systems are essentially standardized within [8]. Leveraging the process reference models for ISMS, IPMS, and BCMS, and drawing upon resources such as the ISO Handbook—The Integrated Use of Management System standards [15]—along with ISO/IEC Directives Part 1 Annex SL [8], a generic process reference model for management systems will be developed. This endeavor aims to facilitate the development of process reference models across various management disciplines and domains.

Step 2: Evaluation of the utilization of process reference models for management systems benchmarking

Drawing from the process reference models for ISMS, IPMS, and BCMS, a study will be conducted to assess how these models can be employed for management systems benchmarking. The premise of this analysis is that benchmarking organizations based solely on sets of best practice controls may not yield substantive insights into an organization's capabilities in managing risks within a specific management discipline. A comparative analysis of management system processes' capability and maturity levels appears to be a more practical approach.

Step 3: Establishing a decision-support system using Fuzzy Inference Systems

We also aim to apply Fuzzy Inference Systems (FISs) in business continuity management to establish a decision-support system. Such a system would help us determine the appropriate BCM actions or decisions regarding different situations. Using FIS in business continuity management will enable us to handle uncertainty inherent in this field, encapsulate the knowledge of experts, and model complex and nonlinear relationships among variables [37], critical for enhancing an organization's preparedness and responsiveness to disruptive events.

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Abbreviations

The following abbreviations are used in this manuscript:

BC	Business Continuity
BCM	Business Continuity Management
BCMS	Business Continuity Management System
BIA	Business Impact Assessment
FIS	Fuzzy Inference Systems
IPMS	Information Privacy Management System
ISO	International Organization for Standardization
ISMS	Information Security Management System
IUMSS	The ISO Handbook "The Integrated Use of Management System Standards"
PDCA	Plan-Do-Check-Act

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